

fi_var_engine_v5.py

Section-by-Section Study Summary

Fixed Income MC VaR Engine | Vasicek / Hull-White | SR 11-7 | Basel III

SECTION 1 - CONFIGURATION

1. Model Selection and Parameters

MODEL_CONFIG selects Vasicek or Hull-White with a single string change. Vasicek sets kappa (mean-reversion speed), theta (long-run mean), sigma (volatility), r0 (initial short rate). Hull-White sets a, sigma, r0 only - theta(t) is time-varying, calibrated from the market yield curve (no-arbitrage model).

2. Simulation Config

n_paths = 10,000 paths, n_steps = 252 steps (1 trading year), horizon = 1 trading day (VaR holding period), seed = 42 for reproducibility.

3. Risk Config

Confidence level = 99% (alpha = 0.01), holding period = 1 day. Matches Basel III regulatory VaR standard.

4. Yield Curve

FRED API Key fetches live yield curve data. Without a key, falls back to hardcoded curve: 10 tenors from 0.25yr to 30yr, yields from 4.30% to 5.20% (upward sloping).

5. Stress Scenarios and Backtest Config

6 stress scenarios: Parallel Up +200bps, Parallel Down -100bps, Steepener (short end down, long end up), Flatteners (short end up, long end down), Credit Spread Widen +50bps, Recession Flight to Quality -15bps. Backtest window: 250 days, alpha = 1%.

SECTION 2 - NELSON-SIEGEL YIELD CURVE

1. Model Formula and Economic Meaning

Nelson-Siegel (1987): $y(\tau) = B_0 + B_1 * f_1(\tau) + B_2 * f_2(\tau)$. B_0 = long-end level (long-run interest rate level), B_1 = slope (short-end loading), B_2 = curvature / hump (mid-curve shape), λ = decay factor. All four parameters have clear economic meaning, which is a key advantage over pure mathematical interpolation.

2. Parameter Estimation

Nelder-Mead numerical optimization minimizes fitting error: $\sum (y_{NS} - y_{market})^2$. Initial guess: B_0 = long-end yield, B_1 = short-long spread, B_2 = small hump, $\lambda = 2$.

3. Advantages over Linear Interpolation

Smooth and kink-free: no jumps at knot points. Stable extrapolation: reasonable estimates beyond the longest tenor (30yr). No-arbitrage: continuous smooth curve introduces no arbitrage opportunities. One function ns_interp(t) is reused across MBS pricing, stress tests, and KRD.

4. Output Interpolation Function

Returns callable function ns_interp(t) that outputs the continuously-compounded zero rate for any tenor t. Used directly by MBS pricing, stress testing, and KRD calculation throughout the engine.

SECTION 3 - RATE MODELS

1. Vasicek Exact Discretization

SDE: $dr_t = \kappa * (\theta - r_t) * dt + \sigma * dW_t$ (mean reversion plus random shock). Exact conditional Gaussian discretization (not Euler approximation). One-step std = $\sqrt{[\sigma^2 / (2 * \kappa) * (1 - \exp(-2 * \kappa * dt))]}$. Avoids Euler error amplification under high volatility or long holding periods. Key limitation: Gaussian tails underestimate extreme rate moves.

2. Hull-White Dynamic $\theta(t)$ Calibration

SDE: $dr_t = (\theta(t) - a * r_t) * dt + \sigma * dW_t$. $\theta(t)$ is time-varying, determined by the market forward rate curve: $\theta(t) = a * f(0, t) + df/dt + \sigma^2 / (2a) * (1 - \exp(-2 * a * t))$. Achieves exact fit to the initial term structure (no-arbitrage property). Uses Euler-Maruyama discretization. Key difference from Vasicek: θ is a function, not a constant - no fixed long-run mean.

3. Antithetic Variates - Variance Reduction

Generates paired random numbers Z and $-Z$ (half paths use Z , other half use $-Z$). Halves Monte Carlo variance at zero extra computational cost. With 10,000 paths, only 5,000 random numbers are actually generated.

4. `simulate_horizon()` - Fast Path

Directly samples the short rate at the VaR horizon (1 day) from the exact conditional distribution, bypassing all 252 intermediate simulation steps. Speed improvement: approximately 250x faster. Vasicek: uses exact Gaussian distribution. Hull-White: uses Euler one-step approximation (time-varying $\theta(t)$ has no closed-form exact sampler).

5. `discount_factor_batch()` - Vectorized Pricing

Computes ZCB prices for multiple maturities in one call, returning an ($n_times \times n_paths$) matrix. Replaces n_times separate function calls, eliminating Python overhead. Vasicek closed form: $P(t, T) = A(\tau) * \exp(-B(\tau) * r)$. HW closed form: $P(t, T) = P_mkt(0, T) / P_mkt(0, t) * \exp(-B * (r - f(0, t)) + adj)$.

SECTION 4 - DATA PIPELINE

1. Position Data (Simulated SQL)

12 product positions: UST 2yr and 10yr, Corp IG 3yr and 7yr, Corp HY 5yr, MBS Pool A and B (with wac = weighted average coupon, cpr = baseline prepayment rate), ABS Auto, ABS Card, CLO AAA, CLO BBB. Each record contains: notional, coupon, maturity, credit_spread, DV01, effective_duration.

2. Historical P&L (Synthetic Data)

250-day historical P&L = Normal($\sigma=185K$) + t-distribution($df=5$, $\sigma=42K$) mixture. Simulates leptokurtosis (fat tails, high peak) of real P&L distributions. Important limitation: synthetic data reduces Backtesting statistical validity. Production systems must use real historical P&L and handle missing data, holidays, and outliers.

SECTION 5 - RISK ENGINE

1. Full Repricing

Every MC path reprices the entire portfolio (not Delta approximation). Plain bonds: Vasicek/HW analytical ZCB formula. ABS: effective duration linear approximation ($dP = -D * dr * P$). MBS: PSA monthly cash flow discounting. $P\&L = P1(r_horizon) - P0(r0)$.

2. PSA Prepayment Model for MBS

Rate-sensitive prepayment: $CPR_eff = CPR_base * (1 + \beta_{refi} * (r0 - r))$, with $\beta_{refi} = 8.0$. Includes PSA ramp-up period (months 1-30 ramp linearly to full speed). $SMM = 1 - (1 - CPR)^{(1/12)}$. Monthly cash flows (interest + scheduled principal + prepayment) are discounted. When rates fall, CPR rises (refinancing wave)

causing negative convexity (price appreciation is capped).

3. VaR and CVaR Calculation

$\text{VaR}(99\%) = -\text{percentile}(\text{P\&L}, 1\%)$. $\text{CVaR}(99\%) = -\text{mean}(\text{P\&L} \mid \text{P\&L} < -\text{VaR})$. CVaR is always $\geq$ VaR. CVaR is a Coherent risk measure (satisfies subadditivity). VaR is not. Basel FRTB has shifted from 99% VaR to 97.5% ES (Expected Shortfall = CVaR).

4. Stress Testing (6 Scenarios)

Each scenario: shock yield curve, refit Nelson-Siegel, reprice full portfolio, output P&L impact. Largest loss: Credit Spread Widen -\$21.9M (far exceeds Parallel Up -\$9.8M). Key conclusion: portfolio risk is dominated by credit spread exposure, not pure interest rate risk. Limitation: scenarios are applied individually - combined shocks (rate up AND spread widen, as in 2008) are not modeled.

5. Key Rate Duration (KRD)

Applies isolated +1bp shock at each tenor (all other tenors unchanged). $\text{KR01} = -(\text{P}_{\text{bumped}} - \text{P}_{\text{base}})$. $\text{KRD} = \text{KR01} / (\text{P}_{\text{base}} * 0.0001)$. Captures twist risk (short vs long end) and butterfly risk (short/long vs mid), which a single DV01 (parallel shift only) cannot distinguish.

SECTION 6 - MODEL VALIDATION

1. Kupiec POF Test - Unconditional Coverage

H0: actual exception rate $\hat{p}$ = model expected exception rate α (1%). $\text{LR} = -2 * [N * \log(\alpha / \hat{p}) + (T-N) * \log((1-\alpha)/(1-\hat{p}))]$, follows chi-squared(1) distribution. Reject H0 if $p < 0.05$ (FAIL). Current result: exceptions = 0, LR = infinity, $p = 0.0000$, FAIL. Reason: model is too conservative - historical P&L never breached VaR. Root cause: synthetic P&L distribution does not perfectly match MC model distribution.

2. Christoffersen CC Test - Independence

Extends Kupiec by testing whether exceptions are time-independent. Builds transition matrix: n_{00} , n_{01} , n_{10} , n_{11} (0=no exception, 1=exception, yesterday vs today). Large n_{11} means exceptions cluster (volatility clustering) - model underestimates risk persistence. $\text{LR}_{\text{CC}} = \text{LR}_{\text{UC}} + \text{LR}_{\text{IND}}$, follows chi-squared(2). Interview key point: Kupiec checks quantity only; Christoffersen also checks time-series structure.

3. Basel Traffic Light

Exception count in 250-day window determines zone: 0-4 exceptions: Green zone (capital multiplier $k = 3.0x$, no action required). 5-9 exceptions: Yellow zone ($k = 3.4x$, enhanced monitoring). 10+ exceptions: Red zone ($k = 4.0x$, immediate review and capital add-on). These thresholds directly affect regulatory capital requirements - must memorize.

4. Historical Simulation Benchmark

HS VaR/CVaR = empirical quantile of historical P&L (no model assumptions). Compared against MC VaR to quantify model deviation ($\Delta \text{VaR}\%$). HS advantage: captures real tail distribution without distributional assumptions. HS disadvantage: backward-looking, unsuitable for new products or scenarios not present in history.

5. Sensitivity Analysis

Perturbs σ ($\times 0.75$ and $\times 1.25$) and a ($\times 0.50$ and $\times 1.50$), outputs $\Delta \text{VaR}\%$ and $\Delta \text{CVaR}\%$. Results: $\sigma \times 1.25$ gives +26.8%, $\sigma \times 0.75$ gives -27.2%. $a \times 0.50$ gives +19.1%, $a \times 1.50$ gives -14.3%. Conclusion: σ impact on VaR ($\sim 27\%$) is roughly double that of a ($\sim 14-19\%$). Volatility is the most sensitive parameter - its estimation accuracy is critical to VaR reliability.

SECTION 7 - CHARTS

1. P&L Distribution

Histogram with normal fit overlay, vertical lines marking VaR and CVaR. CVaR line is to the left of VaR (larger loss). Actual distribution has slightly heavier left tail than normal, reflecting the limitation of the Gaussian

assumption in underestimating extreme event frequency.

2. Simulated Rate Paths

200 sample paths plus Mean Path plus 5th-95th percentile band (fan shape). Hull-White Mean Path direction is driven by market forward curve, not a fixed target mean. The 5th-95th band widens over time but at a decreasing rate (mean reversion prevents unlimited divergence).

3. Stress Test Bar Chart

Horizontal bar chart of P&L impact for all 6 scenarios. Red = loss, Green = gain. Credit Spread Widen at -\$21.9M is by far the largest loss. Recession Flight to Quality at +\$8.5M is the largest gain (rates fall, long-duration portfolio benefits). Visually confirms credit spread is the dominant risk.

4. Sensitivity Delta VaR Bar Chart

Horizontal bars showing delta VaR% for each parameter perturbation. $\sigma \times 1.25$: +26.8%. $\sigma \times 0.75$: -27.2%. $\alpha \times 0.50$: +19.1%. $\alpha \times 1.50$: -14.3%. σ impact is approximately twice that of α .

5. alpha x sigma VaR Heatmap

2D heatmap: 7 volatility multipliers (y-axis) x 5 confidence levels (x-axis). Bottom-left corner ($\sigma \times 0.6$, 90%): \$0.17M (darkest green, lowest risk). Top-right corner ($\sigma \times 1.5$, 99.5%): \$1.01M (darkest red, highest risk). Baseline ($\sigma \times 1.0$, 99%): \$0.59M. Moving along y-axis (changing σ) has greater impact than along x-axis (changing α).

6. KRD Profile

Two-panel bar chart: KRD in years (left) and KR01 in dollars (right) by tenor. Shows where duration risk is concentrated across the yield curve. Supplements DV01 by revealing curve shape exposures: twist (short end vs long end) and butterfly (short/long vs mid).

SECTION 8 - SR 11-7 DOCUMENTATION

1. Auto-Generated Compliance Document

All calculation results from Sections 1-7 are inserted into an SR 11-7 template, outputting `model_risk_documentation.md`. The 580 lines of code are entirely string templates with no new mathematical logic. No need to study the code itself - read the generated .md file instead.

2. Document Structure (Interview Must-Know)

1. Executive Summary. 2. Model description and parameter table (Vasicek/HW formulas, discretization). 3. Assumptions and limitations list (single-factor, Gaussian tails, synthetic data, simplified MBS). 4. Three-pillar validation results (Backtesting / HS Benchmark / Sensitivity). 5. Governance framework (Three Lines of Defense, Tier 1 classification, annual review, Model Inventory). 6. Document Control (Model ID, Owner, Validator, Next Review Date).

3. Three Lines of Defense and Tier Classification

Line 1: Development team (Front Office) - builds the model, cannot self-validate. Line 2: Model Risk Management (MRM) team - independent validation, required by SR 11-7 to be separate from Line 1. This is the role you are applying for. Line 3: Internal Audit - reviews the entire process, reports to the Board. Tier 1 (high impact, directly affects regulatory capital): full validation annually. Tier 2 (medium impact): every two years. Tier 3 (low impact): every three years. This project is classified Tier 1.

4. Attestation Statement

The document explicitly states the model has not yet received formal MRM validation. Production deployment requires: independent validation, CRO approval, Model Inventory registration, and SR 11-7 compliance sign-off. This demonstrates understanding of the complete MRM workflow - a key differentiator from a standard quant project.